

ICP 01. The Itching Pregnant Patient

Companion reading handout for simultaneous listening.

Audio course on intrahepatic cholestasis of pregnancy. Episode one. The itching pregnant patient.

Source Base

This episode is based on the Israeli position paper on intrahepatic cholestasis of pregnancy, Gabbe chapters on hepatic disease, maternal physiology, skin disease, and normal pregnancy laboratory values, and the SMFM consult on intrahepatic cholestasis of pregnancy.

Opening Frame

Let us begin with the central clinical idea.

Intrahepatic cholestasis of pregnancy, or ICP, is not diagnosed because a pregnant patient is itchy. It is suspected because the pattern of itching behaves like cholestasis, and it is supported by bile acids.

That distinction is important. Itching is common in pregnancy. ICP is not. If we call every itch ICP, we overdiagnose. If we dismiss classic itching because the first bile acids are normal, we miss disease that is still evolving.

So the first job of the resident is not to memorize a threshold. The first job is to recognize the phenotype.

The classic patient is in the late second trimester or the third trimester. The Israeli position paper notes that about eighty percent of women develop the disorder after thirty weeks. She has pruritus, usually without a primary rash. The itch may be generalized, but it is classically worse on the palms and soles, and often worse at night.

That nighttime feature matters. A patient may say, I cannot sleep. My hands and feet are burning. I do not have a rash, but I am scratching until I make marks.

Those scratch marks are not a primary rash. They are excoriations. This is one of the most useful bedside distinctions. In ICP, the itch comes first and the skin damage comes second. In many pregnancy dermatoses, the lesion comes first and the itch is produced by the lesion.

Pathophysiology

Now step back and ask, why does this happen?

In ICP, bile acids are not being handled normally by the maternal liver. They accumulate in the maternal circulation and in tissues, including the skin. Gabbe describes the cardinal symptom as pruritus from cholestasis and accumulated bile salts in the dermis. That is why the patient is miserable even though the skin exam may be deceptively unimpressive.

Jaundice can occur, but it is not common. The Israeli position paper and Gabbe both place jaundice around ten percent. When jaundice is present, you should not panic, but you should widen your lens. Jaundice means bilirubin is now part of the story. It raises the stakes for evaluating severity, alternative hepatobiliary disease, and coagulation or vitamin K issues.

The natural history also helps. ICP is usually pregnancy-limited. It appears late, worsens as pregnancy continues, and usually improves rapidly after delivery. That is because delivery removes the hormonal and placental environment that is stressing bile transport.

But pregnancy-limited does not mean trivial. For the mother, the disease can cause severe sleep loss, anxiety, distress, jaundice, and vitamin K concerns. For the fetus, the risk story is completely different, and we will build that in later episodes.

Risk Factors

Risk factors sharpen your pretest probability.

A history of ICP in a previous pregnancy matters a lot. The Israeli position paper reports recurrence up to ninety percent, often earlier in later pregnancies. Family history matters, because the disease has a genetic component. Multifetal pregnancy and IVF pregnancy matter, probably because placental and hormonal load are higher. Maternal age over thirty-five and background hepatobiliary disease, such as gallstones or cholecystitis, also increase suspicion.

These risk factors do not diagnose ICP. They tell you how hard to keep looking when the story sounds right.

Diagnosis

Now let us move to diagnosis.

The core tests are total bile acids and liver transaminases. In spoken clinical language, we usually say AST and ALT. These tests do different jobs. Bile acids are

central for diagnosis and fetal-risk stratification. AST and ALT tell you that the maternal liver is irritated, but they are not the main fetal-risk biomarker.

A useful first panel for a classic patient includes bile acids, AST, ALT, bilirubin, and enough basic maternal assessment to avoid missing a dangerous mimic. If the presentation is severe or atypical, add platelets, creatinine, urine protein assessment, and coagulation studies. If there is jaundice, systemic illness, pain, fever, or persistent postpartum abnormality, think about viral hepatitis, biliary obstruction, medication injury, and other liver disease.

Normal Pregnancy Trap

There is one normal pregnancy trap worth emphasizing.

Alkaline phosphatase rises in pregnancy because the placenta produces alkaline phosphatase. So an elevated alkaline phosphatase is much less useful than it would be outside pregnancy. By contrast, transaminases do not normally rise in the same way. If AST or ALT are elevated, take them seriously.

Diagnostic Bile Acid Threshold

Now, what bile acid value supports the diagnosis?

The common diagnostic threshold is above ten micromoles per liter in a symptomatic patient. Gabbe's normal pregnancy laboratory table places elevated total bile acids above ten. The Israeli position paper uses symptoms plus bile acids above ten micromoles per liter. SMFM also describes total bile acids above ten as a commonly used diagnostic value.

But do not treat ten as a magic wall. It is a clinical threshold, not a complete disease model. A patient with classic pruritus and bile acids of eight may be early in the disease. A patient with vague abdominal itching and bile acids of eleven may need context before you make a confident diagnosis.

Pruritus Can Precede Abnormal Labs

This brings us to one of the most important practical points in the whole topic.

Pruritus can precede abnormal bile acids.

The Israeli position paper states this clearly. SMFM also emphasizes that itching can come before the rise in serum bile acids by several weeks. So if a patient has classic symptoms and normal initial bile acids, the correct response is not, this is impossible. The correct response is, this is not confirmed yet. Repeat bile acids and transaminases if the symptoms persist.

This matters because a single normal lab result can create false closure. The patient keeps itching. The disease evolves. The clinician stops looking. That is exactly the wrong pattern.

Fasting and Nonfasting Bile Acids

Now let us talk about fasting and nonfasting bile acids.

The Israeli position paper says fasting is desirable but not essential. SMFM states that random bile acids can be used, even though bile acids vary after meals. Ovadia's meta-analysis included mixed fasting and postprandial sampling, and the median values were similar in the restricted analysis.

The practical approach is simple. If fasting bile acids are easy, they are clean. If nonfasting bile acids are what you have, do not delay care. The fetus does not wait for perfect sampling conditions. What matters is that you interpret the number in context, repeat testing when needed, and follow the peak bile acid level over time.

Bedside Pattern Recognition

Now let us anchor the resident's first encounter with a patient.

Imagine a woman at thirty-three weeks. She says, I am itching everywhere, but especially my palms and soles. It is worse at night. There is no urticarial eruption, no vesicles, no abdominal striae eruption, no fever, no right upper quadrant pain. Her blood pressure is normal. Platelets are normal. AST and ALT are mildly elevated. Bile acids are twenty-six.

This is a typical ICP pattern. She needs treatment for symptoms, counseling, repeat labs, fetal surveillance at the appropriate gestational age, and delivery planning by peak bile acids.

Now change the case.

She is thirty-three weeks with itching, epigastric pain, blood pressure of 160 over 110, platelets of eighty thousand, and AST twice normal. That is not a simple ICP story. That is preeclampsia with severe features or HELLP until proven otherwise.

Change it again.

She is thirty-three weeks with itching and a tense bullous rash around the umbilicus. That is not typical ICP. Think pemphigoid gestationis and dermatology involvement.

Change it again.

She is ten weeks with severe vomiting, weight loss, ketonuria, and liver-test abnormalities. That is not the classic time course of ICP. Think hyperemesis gravidarum and its differential.

This is how ICP should be taught. Not as one test. As a pattern.

Summary

Before we end, let us summarize the first-episode framework.

First, ICP usually presents in late pregnancy with pruritus, often palms and soles, often worse at night, and usually without a primary rash.

Second, prior ICP, family history, multifetal pregnancy, IVF, older maternal age, and hepatobiliary disease raise suspicion, but they do not replace bile acids.

Third, bile acids are central. In a symptomatic patient, values above ten micromoles per liter support the diagnosis. But normal initial bile acids do not permanently exclude ICP if symptoms persist.

Fourth, AST and ALT support hepatic involvement, but alkaline phosphatase is weak in pregnancy because it normally rises from the placenta.

Fifth, the skin exam matters. ICP has itch first and excoriations second. A primary eruption moves you toward pregnancy dermatoses or another skin disease.

And the first clinical memory hook is this.

Itch first. Rash absent. Bile acids decide.

End of episode one.